

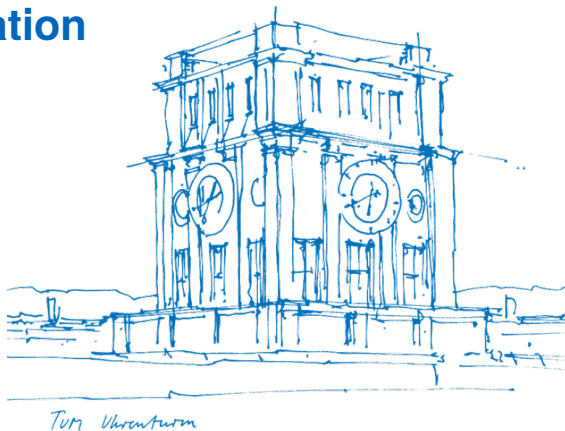
Agentic AI for Code Generation

Seminar Report — SS26

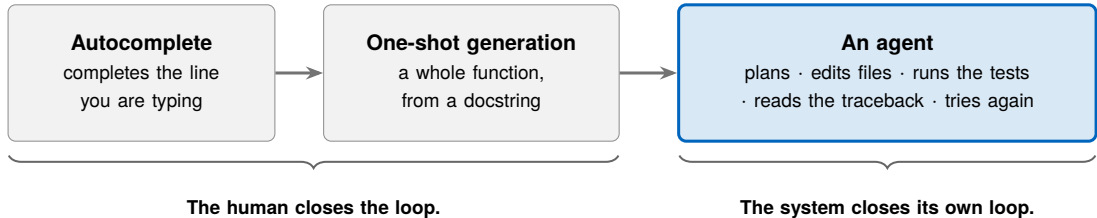
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July 19, 2026

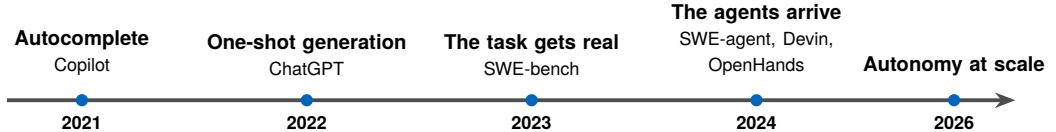


What is agentic AI for code generation?



Prose in; a patch out.

Three years, three eras

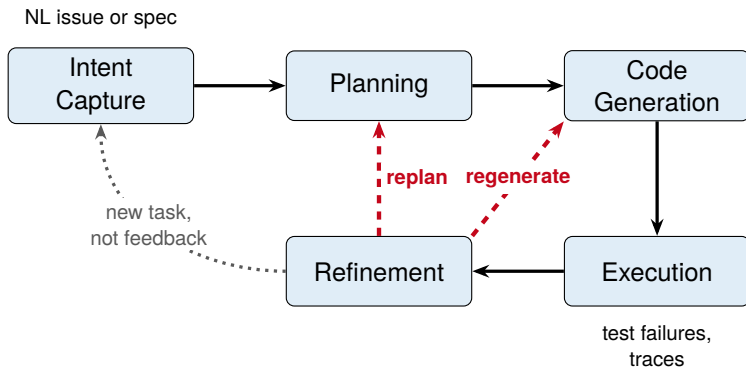


under 2% → **over 70%**

of real GitHub issues resolved autonomously [10, 20]

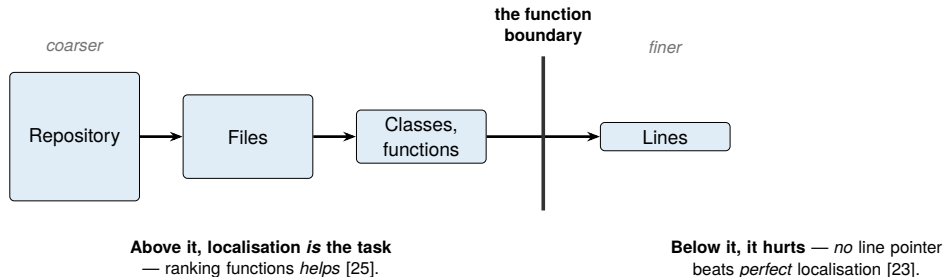
(two different rulers — defined in a few slides)

The agent pipeline



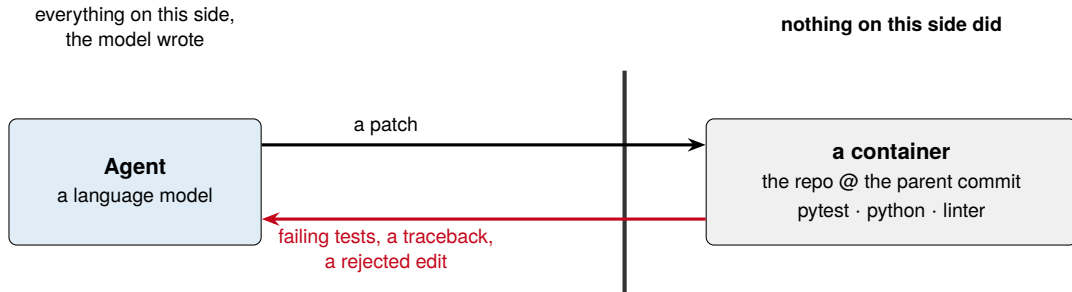
Five stages — **one vocabulary for every system in this talk.**

Planning is localisation, not design



Reasoning about *where* the code is — not *how* to write it.

Execution: the code meets the environment



Not a score — *a symptom*, still to be read.

Refinement: what to do with the signal

- **Repair from the trace** [9]
- **Explain your own code** [3] — *only where tests exist*
- **Sample N , rerank with a trained critic** [15]

Agents improve their own code.

What SWE-bench measures

One instance [10] = a real merged PR: closed an issue + touched the tests.

The agent sees the repo before the merge + the issue; **returns** a patch.

Graded by running the tests — the bug fixed, nothing broken.

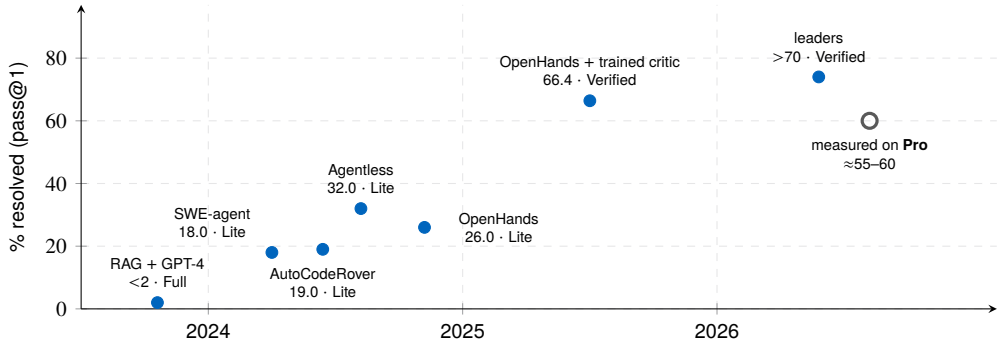
Resolve rate = both, on the first patch.
Every number from here on is this one.

One benchmark, four splits:

- **Full** — 2,294, every scraped PR
- **Lite** — 300, single-file patches
- **Verified** — 500, human-screened
- **Pro** — 1,865, held-out repos

Rates are not comparable across splits. Verified is the most *curated* split, not the hardest.

Three years, in numbers



Splits are **not comparable** — read the *climb*, not the deltas. [10, 24, 25, 22, 21, 15, 20, 16]

No scaffold wins twice

	SWE-agent custom ACI	OpenHands code-as-action	mini-SWE-agent ~100 lines of Python	spread
GPT-5-mini	38.8	38.6	51.8	13.2
Qwen3 Coder 480B	36.6	54.0	40.2	17.4
DeepSeek R1	22.6	18.0	11.0	11.6

Resolve rate (pass@1) on SWE-bench Verified, April 2026 [18, 19]. ACI = agent–computer interface: the command set the model is handed.

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- Same model, same benchmark: the scaffold alone is worth **11–17 points**.
- **No scaffold wins twice** — the best scaffold *depends on the model*.

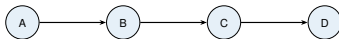
- In three years: under 2 % → **over 70 %**.
- **What changed is the loop** — the system runs its own code, and reads what comes back.
- On the harder, held-out split: $\approx 60\%$ [16]. *The climb is not over.*

Backup: every refinement mechanism, sorted

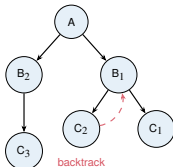
Mechanism	Closed by	Signal	Characteristic failure
Self-Refine [12]	S	Model critiques own output	Plateaus; nothing can contradict the generator
Self-debug, no tests [3]	S	Self-generated explanation	Explains the bug as intended behaviour
Self-debug, with tests [3]	O+S	Test verdict + model diagnosis	Bounded by test coverage
Reflexion [17]	O or P	Task reward where one exists; on HumanEval, self-generated tests	Filed as external, but its code result rests on a proxy
SelfEvolve [9]	O	Interpreter errors, traces	Silent (non-crashing) semantic bugs
D4C [23]	O	Failing tests + full function	Plausible-but-incorrect patches
CRITIC [8]	O	Self-critique routed <i>through</i> an external tool	Needs a tool covering the error class
Agentless validate [22]	O+P	Regression tests + generated reproduction tests	The proxy half inherits the model's misreading
Best-of- N + critic [15]	O+P	Executed rollouts ranked by a trained critic	Critic is a distilled proxy; cost scales with N

Backup: how plans are shaped

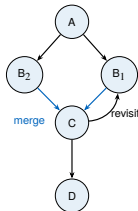
Chain of Thought



Tree of Thoughts



Graph of Thoughts

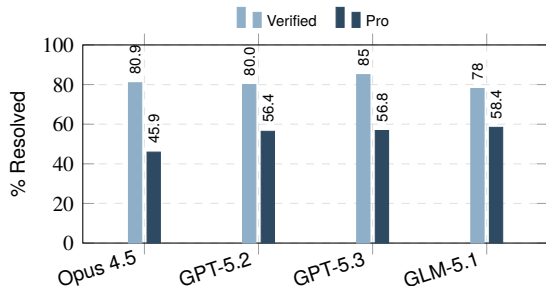


The evaluator, not the branching, makes the structure useful — and it is what a repository-scale agent finds hardest to supply. *What is a half-finished patch worth, before any test has run?* SWE-Search [1] builds one (MCTS + LLM value function, +23% relative) — and must invoke a model at every node.

Backup: the numbers behind the stage walk

- **Planning, above the function boundary** [25]. AutoCodeRover adds spectrum-based fault localisation to rank *functions*: **+3 points** on SWE-bench Lite.
- **Planning, below it** [23]. D4C hands the model the failing test and the *whole function*, and gives it *no line-level pointer at all* — beating systems given perfect localisation by **10 %**. Below the function boundary, a pointer fights the model's infilling prior.
- **Refinement, with an explanation** [3]. Self-debug: up to **+12 %** — but only where a test suite exists to run against.
- **Refinement, with a trained critic** [15]. OpenHands on Verified, best-of- N reranked: 60.6 → **66.4 %**. (On the deck: the numbers timeline.)

Backup: the benchmark, in more detail



The *same weights* drop $\approx 20\text{--}35$ points on the contamination-resistant SWE-bench Pro [4].

Read only the floor. Pro is harder *as well as* fresher, and the column mixes harnesses — so the drop *size* is not a measurement of memorisation.

91 % of Verified is work a developer finishes *within the hour* [6]. A system can gain points while solving no additional hard task.

July 2026: the ruler itself is under audit. OpenAI flags $\sim 30\%$ of Pro's public tasks as broken and retracts its February recommendation to use it [14]. Quote *measured* Pro numbers, with that caveat attached.

Backup: context and cost

- **Longer windows do not dissolve the bottleneck.** Retaining full *dependency* context works best; padding with loosely related code *degrades* correctness [13].
- **Context explosion and semantic drift.** Appending every observation buries the critical early information [11].
- **Cost is quadratic in turns.** The whole prefix is re-sent each turn, so tokens grow as $O(n^2)$ [7]. A property of the protocol, not an empirical finding; prompt caching cuts the constant, not the growth rate.
- **Resolve rate decouples from compute** [5]. Agentless leads on resolve rate (48 % with Qwen3-32B) at 83.1 API calls and 209.4 s per issue. The frontier is accuracy *per dollar*.
- **And verification is the human bottleneck.** In an RCT, experienced developers were 19 % *slower* with early-2025 AI tools — while believing they had been 20 % faster [2].

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